

PROJECT
RED SKY

Built into the Core, Enforced Across Every App and Handset

Advising New Telecom Ltd (NTL) Executive Management

GAPS & CASE STUDY

Check backup slides for citations of all case studies



Fatema – Student

- **Parcel leak:** a delivery rider copies her number from the parcel sticker and harasses her at night
- **App privacy can't help:** couriers call her normal SIM directly.



Fahim – 10 Years Old

- **Shared phone:** he plays games on his mother's handset and is profiled as an adult.
- **Youth ban:** profiling minors breaks PDPA Sec 9 & Def 2(13)



Tanvir – High-ARPU Executive

- **Number takeover:** when his number is recycled or swapped, a stranger can receive his bank OTPs.
- **Silent airtime drain:** third-party billing drives about 9M dispute calls a year (est.).



Kashem Bhai – Farmer

- **Open ledger:** his number sits in a paper khata at 250,000+ recharge stalls.
- **Button phone:** no app or web page, so no privacy settings. 20M+ users are in his position.



Statutory Conflict

- **Opposing Mandates:** PDPA Sec 18 erasure clashes with BTRC Sec 97 retention.
- **Compliance Deadlock:** A 24-month legal hold blocks standard database deletion.



Enterprise Data & AI Egress

- **Dark Lakehouses:** Unmapped Hadoop/BSS stores lack automated lineage.
- **Cloud AI Egress**: Telemetry leaks cross-border to foreign AI vendors (Sec 29–30).

Our Approach

PRIVACY BUILT INTO THE SIM. ENFORCED AT THE CORE. TRUSTED BY DESIGN.

Network-layer control can't be uninstalled, works with no data connection, and reaches every subscriber—an app can't claim any of the three.

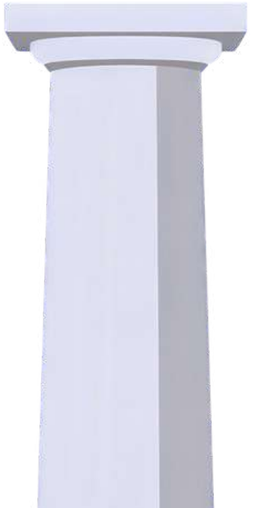
ENTERPRISE GOVERNANCE

Lakehouse Lineage, Sovereign AI Gateway & Egress Control



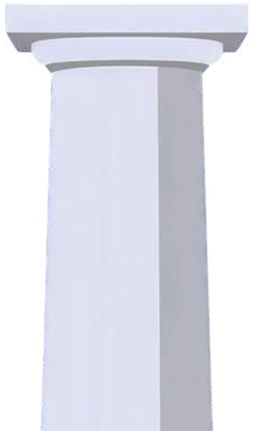
CONSUMER PROTECTION

Zero-Trust Edge Safety, Identity Masking & Balance Defense



REGULATORY TRUST & IDENTITY

The Clean Wall, GSMA CAMARA APIs & Statutory RACI



NETWORK-LAYER CONSUMER PROTECTION

Stealth Number (ছদ্মবেশী মোড)

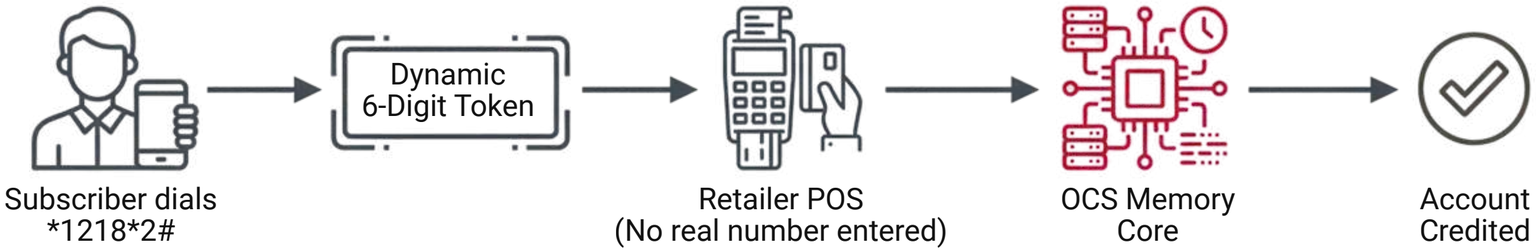
Stops couriers from keeping your real number, while courts can still trace it.



1. Request a temporary number.
2. Provide it to the rider.
3. Calls are forwarded to your real phone.
4. Expires in 24h, on delivery, or manually
5. Unmasks for High Court warrants

Blind Flexiload (The No-Khata Mandate)

Ends paper ledgers. The shopkeeper types a one-time code, never your number.
The token resolves in memory via the OCS charging core.



1. Generate a 6-digit code.
2. Read it to the shopkeeper.
3. Shopkeeper enters the code.
4. Expires fast; no numbers logged.

Proven Precedents: Safaricom Kenya (298k agent ledger ban) • Vodafone Sakhi (India tokens)



DIGITAL INCLUSION & YOUTH PROTECTION (PDPA SEC 21 & 18)

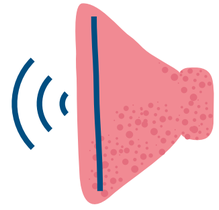
CONSENT FOR EVERY HANDSET



AGE PROOF WITHOUT ID (The "VIP wristband")

The SIM proves it is genuine and a secure network vault confirms "Over 18: YES". Platforms never see a National ID, birthdate or photo.

Keypad & Dialect Voice Consent



Every PDPA right on a button phone, with no internet. Dial *1218# to view, correct, withdraw, port or erase your data. Or call 1218 and say "হ্যাঁ/না" in your dialect (4 regional dialects). You get a signed SMS receipt, and a yearly privacy check-up reminds you to review your choices.



Guardian Child-Shield

A parent dials *12184# and confirms with the line PIN. Ad tracking and behavior profiling switch off for that line, and the parent can reverse it any time. Supports PDPA Sec 21 and 35.

YOUR BALANCE AND YOUR NAME, LOCKED

Airtime Vault, bank SIM-swap checks and NID Watch protect what is yours.



NID WATCH
(*12185#)

- Instant alert if a SIM is registered on your NID; one key locks it.
- **Clean recycling**: banks are alerted to delink old accounts before a number is reissued.
- **Fresh swap**: a free new number, with your balance moved, if your number is exposed in a confirmed breach.

Airtime Vault
(*1218*1#)

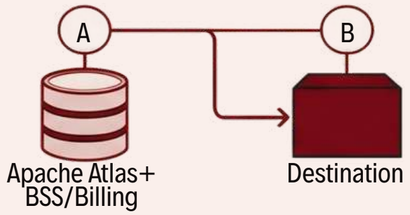
- **Balance lock:** no third-party charge touches your airtime without your USSD PIN.
- **Automatic refund:** if one slips through, you are refunded and NTL recovers it from the developer's 14-day escrow.
- **Sandboxed developers:** partners run in a sandbox and never see your number.

BANK SIM-SWAP CHECK
(GSMA CAMARA)

- Banks ask NTL "was this SIM swapped recently?" before money moves.
- **True or false only:** no call records or identity leave NTL.
- **New revenue:** banks pay per check. Operators such as Dialog Axiata already offer this.

Knowing Where Every Byte Goes

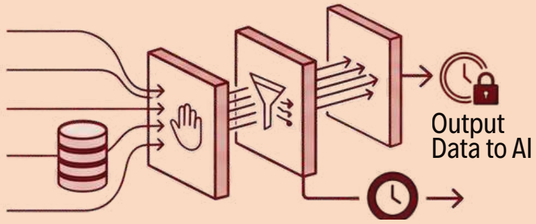
Automated data lineage, an AI gateway, and egress control, on existing infrastructure.



AUTOMATED DATA LINEAGE

(APACHE ATLAS+ICEBERG)

- **OpenLineage:** tracks every dataset across Hadoop, Spark and billing.
- **Vendor registry:** every partner is logged and gets tokens, never raw numbers.
- **Masked Agent Desk**: care agents see masked data and reveal it only with the customer's OTP.



SOVEREIGN AI GATEWAY

(Explainable AI)

- **PII scrubber**: strips numbers, NIDs and names before any prompt reaches an outside model.
- **Injection shield:** keeps system instructions apart from customer text (OWASP LLM01).
- **Explainable decisions:** plain-language Bangla/English action cards.



ZERO-TRUST EGRESS CONTROL

(Outbound DLP & Firewall)

- **Outbound filter:** redacts personal data in traffic leaving the network.
- **Network block:** unapproved foreign connections are dropped.
- **Data stays home:** restricted data and raw radio telemetry stay in Bangladesh (PDPA Sec 29–30); no ML on tower masts.

SIX AI SYSTEMS. SIX RISKS. SIX CONTROLS.

Each AI gets its own safeguards, not a generic policy.

SUPPORT ASSISTANT

OWNER: CMO / CCO

RISK

Prompt injection; customer data reaching third-party AI providers abroad

On-prem gateway strips numbers, NIDs and names before any prompt leaves

PDPA: Minimization · cross-border

FRAUD DETECTION

OWNER: CFO / RISK

RISK

Collecting too much personal and financial data; unfair or unexplained fraud flags

Banks get a true/false SIM-swap answer only; no call records, balances or biometrics leave

PDPA: Minimization

CHURN PREDICTION

OWNER: CMO

RISK

Biased or unexplained decisions about customers; data kept longer than needed

Coarse behavior bands only; no contact graphs; minors excluded; profile wiped on erasure

PDPA: Individual rights · children

MARKETING RECOMMENDATIONS

OWNER: CMO / CCO

RISK

Profiling and tracking children; targeting without consent

Child-Shield lines excluded; adults opt in; contextual offers only

PDPA: Children · lawful processing

NETWORK OPTIMIZATION

OWNER: CTO

RISK

Location and behavior data processed or sent abroad

Trained at regional hubs with added noise; raw data stays in Bangladesh

PDPA: Cross-border · minimization

DOCUMENT PROCESSING

OWNER: CIO

RISK

Sensitive ID data exposed or kept

Read in memory; text verified, image discarded immediately; no photo stored

PDPA: Minimization

ORGANIZATIONAL ACCOUNTABILITY & LEGAL HARMONY

ACCOUNTABILITY FROM TEAM TO BOARD

PDPA requires one named CDO. We operationalized accountability across three operational tiers.



Privacy Champions

- **Embedded Oversight:** Dedicated privacy leads embedded across technical, marketing, and retail business units.
- **Role-Specific Drills:** Developers practice prompt-injection defense; retail agents drill anti-phishing and SIM-swap detection.



Automated Risk Gate

- **Pre-Deployment Auditing:** Mandatory automated checks embedded into CI/CD deployment pipelines for all predictive ML models.
- **Automated Block:** Any model lacking verified lineage tags, consent proof, or failing bias audits is programmatically blocked.



Statutory CDO / DPO

- **Direct Board line:** reports straight to the Board Audit Committee (PDPA Sec 23).
- **Binding sign-off (NTL policy):** high-risk AI cannot launch without the DPO. Holds one of two keys for lawful-interception warrants.

THE CLEAN WALL

Erasure wipes commercial profiles within 24 hours. Call records are kept 24 months in a sealed, air-gapped vault in Jashore, opened only by a High Court warrant.
(Satisfies both PDPA Sec 13(2) and Telecom Act Sec 97)

EIGHT QUESTIONS FROM MANAGEMENT. EIGHT DEFINITIVE ANSWERS.

The case's own questions, answered one by one.

Where is customer data stored?

Mapped automatically across billing, Hadoop and analytics by Apache Atlas, on our private on-premise cloud.

Which apps process sensitive data?

Every system is classified by sensitivity. eKYC reads documents in memory and never stores the photos.

Which AI systems access customer data?

Only through one central gateway that strips identifiers first. All six AI systems have their own controls.

Which vendors access personal data?

Vendors get tokens, not raw numbers. Developers work in sandboxes and are tracked in a vendor registry.

What data leaves Bangladesh?

Restricted citizen data stays on national soil. Network filters block unapproved connections.

Are AI decisions explainable to users?

Yes. Customers get plain Bangla/English SMS cards giving the main reasons for credit, churn or service decisions.

Can we prove compliance in an audit?

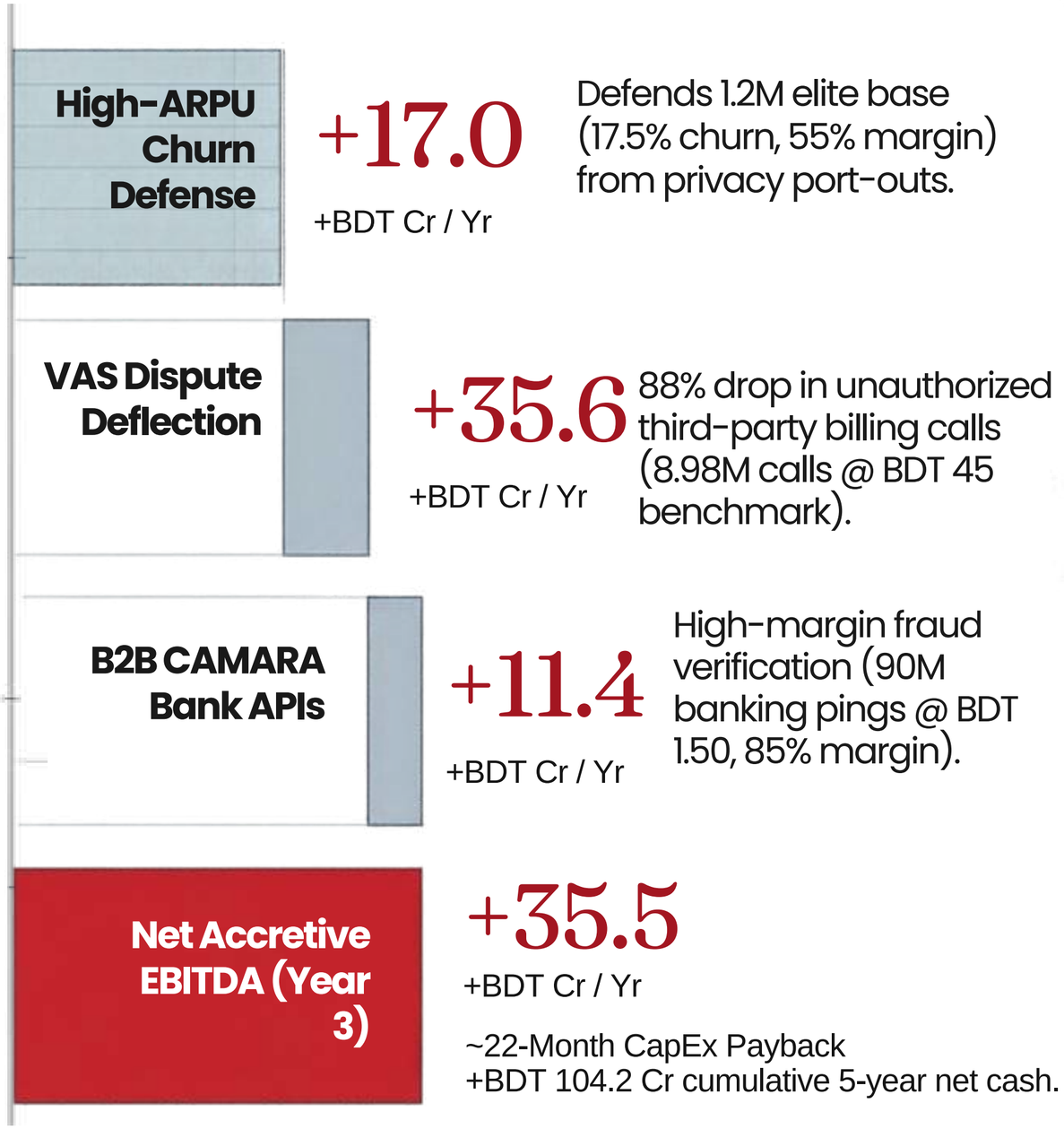
Yes. Tamper-evident logs compile an audit pack for BTRC and the Data Protection Authority in one click.

Who is accountable across business units?

Three tiers: 45 Privacy Champions, automated software safety gates, and an independent DPO reporting to the Board.

MEASURING THE IMPACT

Projected operational and retention returns across 58.6M subscribers



Annual Costs: −28.5 Cr

- −20.2 Cr OpEx & 40 FTEs
- −5.9 Cr Forgone VAS Billing
- −2.4 Cr Refund Buffer

Capital & Stress Test

- 32-Mo Payback at 0% churn
- 29.6 Cr CapEx (Phased Y1/Y2)
- >65 Cr Saved on sunk cloud

Modeled estimate based on NTL's 58.6M Active Subscriber Base

BUILT INTO THE NETWORK.

Privacy as a Network-Layer Utility,
Not an App Feature

Thank you · Team Red Sky



BACKUP SLIDES:

FIELD EVIDENCE

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(Delivery Rider Harassment)

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(Roadside Recharge Ledger)

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B2B FINTECH & LEGAL INFRASTRUCTURE

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SOVEREIGN AI & CYBER DEFENSE

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(Four Phased Milestones)

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(Non-negotiable board covenants)

FIELD EVIDENCE

CASE STUDIES & THE NETWORK MOAT



CASE STUDY: HARASSMENT FOR SHARING OF PHONE NUMBER

The Daily Star

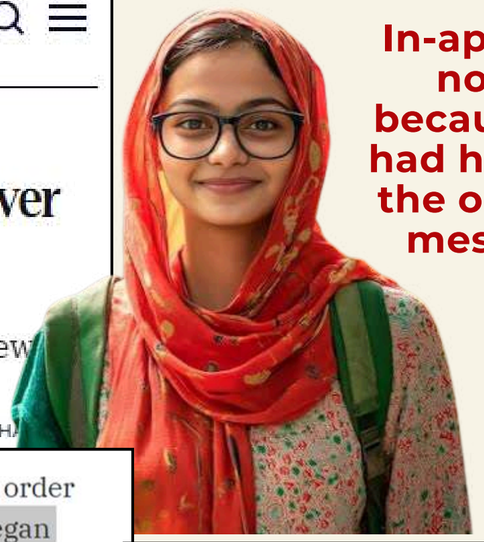
বাংলা

GD filed against food delivery man over harassment of female student

Company says rider terminated after internal review

Published : 8 August 2026 • Updated : 8 August 2026, 23:11

Talking to this newspaper, the student said she placed an order through the platform on August 3. Later that night, she began receiving calls and offensive texts on WhatsApp from an unknown number.



In-app chat privacy does not protect Fatema because the rider already had her real number from the order, and called and messaged her directly

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Bangladesh International Sports Opinion Busines বাংলা

Photo Photo Photo

Bangladesh

Women fall prey to cyber bullying more

Sheikh Sabiha Alam

Published: 29 Mar 2022, 20: 19

A woman lodged a case with Dhaka's Bhashantek Police Station on 15 March. She said, her daughter had ordered food online. After some time, the delivery guy namely [redacted] called her daughter and informed that he won't be delivering her food as there was too much traffic on the road. Her daughter was upset about it and raged at that man a bit.

Since that evening unwanted calls started pouring in her phone. They used objectionable language against her and gave her illicit proposals over phone. After that, marriage proposals started coming in. Later, the victim's family found out that her number along with a groom wanted notice was spread from a Facebook account named 'Alor Pothe'.

Hindustan Times

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Stalkers' delight: Mobile numbers of girls for sale in UP recharge shops

THE DAILY CAMPUS

Latest Education University Admission Updates National International Op-Eds

Home / Crime & Law

Foodpanda Rider Harasses University Student with Indecent Messages After Delivery; GD Filed in Badda

TDC Report | Published: 09 August 2026, 07:39 PM

According to media statements from the victim, she ordered food through Foodpanda at around 10:00 PM on August 3. Later that night, she received calls and explicit messages on WhatsApp from an unknown number. Upon realizing that the caller was the delivery rider who had brought her order, she and her family felt severely vulnerable, prompting

CASE STUDY: ROADSIDE RECHARGE LEDGER EXPLOITATION

THE BUSINESS STANDARD
Wednesday
October 14, 2026

Fraud gang steals bKash money in 30 seconds

CRIME

Nurul Amin
23 December, 2020, 09:55 pm
Last modified: 23 December, 2020, 10:07 pm

Sources said some members of the gang, including [redacted], collected bKash numbers from agent shops. They took photos of the register khata pages of agent shops, secretly, and shared the numbers with other members of the gang.





Kashem Bhai cannot protect his money by changing app settings because his button phone has no app store, and his 11 digits sit exposed in pen on a roadside table

4.1.3 Adherence to paper-based bookkeeping. As mentioned before, FlexiLoad is the primary service offered in FLSs, and this entire process adheres to a paper-based bookkeeping method. When customers want to FlexiLoad to their number, the agent usually notes down the number in his notebook, or at times asks the customers to write down the number by themselves. The amount of money is recorded beside each number and the agent uses this notebook entry to load the required amount to the corresponding number from his own phone.

During his observations, the ethnographer saw that these notebooks are always exposed. When one customer's phone number is being written down, others could easily track and see that number if they want to. The agents do not have any concern about this and they allowed us to take pictures of these notebooks. Figure 2 shows an exposed notebook that contains important information like mobile banking account numbers, phone numbers etc.

Haque et al., ACM ICTD 2020 (DOI: 10.1145/3392561.3394642)

The Financial Express
VOL 33 NO 324 REGD NO DA 1589 Wednesday, 14 October 2026 |

LETTERS TO THE EDITOR

Recycled SIMs and risk to digital identity

October 13, 2026 00:00:00

Tanvir is ready to port his corporate SIM to a competitor because his reissued number receives a stranger's banking alerts, and silent VAS drains his enterprise account balance.

CASE STUDY: WHAT RESEARCH SHOWED

Privacy Vulnerabilities in Public Digital Service Centers in Dhaka, Bangladesh

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at the DSCs in Bangladesh. From interviews with agents and customers, we came to know about several past incidents where female customers' phone numbers were leaked from DSCs to other people, and were used as a means of harassing those customers. In a recent study on online abuse experiences and coping strategies of South Asian women [67], it has also been reported that female customers receive unwanted phone calls after they visit phone recharge (top-up) shops and employees of these shops sell their phone numbers in bulk packages to strangers. Consequently, to

Phone Numbers Leaked & Sold in Bulk from Recharge Stalls

to use that list to call/text people as a part of his election campaign. At times, local goons forcefully take the mobile phone numbers of local girls and harass them over phone. Recently, some of the "Chatro

Harrasement from flexiload shops

interested but he kept harassing me time and again. By some means, I was able to identify the location from which the call had been coming and eventually I realized that it was the neighboring shop from which I regularly FlexiLoad to my number. I confronted the agent and finally he stopped harassing me".

How the Paper Notebook Exposes Numbers to Strangers

4.1.3 Adherence to paper-based bookkeeping. As mentioned before, FlexiLoad is the primary service offered in FLSs, and this entire process adheres to a paper-based bookkeeping method. When customers want to FlexiLoad to their number, the agent usually notes down the number in his notebook, or at times asks the customers to write down the number by themselves. The amount of money is recorded beside each number and the agent uses this notebook entry to load the required amount to the corresponding number from his own phone.

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Actual Photo of the Exposed Khata

Figure 2: An exposed notebook inside a FLS.

CASE STUDY: SHARED-HANDSET DILEMMA & & CHILD DATA PROTECTION

From the competition case file:

- **Prohibitions on Tracking Children**: Organizations are strictly prohibited from conducting targeted advertising, tracking, profiling, or behavioral monitoring directed specifically at children (minors under 18)



Because the SIM belongs to his mother, commercial AI profiles 10-year-old Fahim as an adult consumer,and his data is processed without a guardian's consent, which PDPA Section 9 requires.

ব্যক্তিগত উপাত্ত সুরক্ষা আইন, ২০২৬

(২০২৬ সনের ৬৩ নং আইন)

শিশু বা সম্মতি প্রদানে সক্ষম নহে এমন ব্যক্তিসম্পর্কিত ব্যক্তিগত উপাত্ত প্রক্রিয়াকরণ	৯। (১) শিশু বা সম্মতি প্রদানে সক্ষম নহে এমন ব্যক্তির পক্ষে তাহার পিতামাতা বা আইনগত অভিভাবক বা সিদ্ধান্ত প্রদান করিতে ক্ষমতাপ্রাপ্ত কোনো ব্যক্তির সম্মতি গ্রহণক্রমে, উপাত্ত-জিম্মাদার প্রবিধান দ্বারা নির্ধারিত পদ্ধতিতে, কোনো শিশু বা সম্মতি প্রদানে সক্ষম নহে এমন ব্যক্তির ব্যক্তিগত উপাত্ত সংগ্রহ বা প্রক্রিয়া করিতে পারিবেন। (২) শিশু বা সম্মতি প্রদানে সক্ষম নহে এমন ব্যক্তির অধিকার ও স্বার্থ যাহাতে সুরক্ষিত থাকে বা ক্ষুণ্ন না হয় সেই দিকে লক্ষ্য রাখিয়া শিশু বা সম্মতি প্রদানে সক্ষম নহে এমন ব্যক্তির ব্যক্তিগত উপাত্ত প্রক্রিয়া করিতে হইবে।

(১৩) “পরিলেখা (profiling)” অর্থ কোনো ব্যক্তির ব্যক্তিগত তথ্যের স্বয়ংক্রিয় প্রক্রিয়াকরণ যাহা কর্মক্ষেত্রে ব্যক্তির কর্মক্ষমতা, অর্থনৈতিক পরিস্থিতি, স্বাস্থ্য, ব্যক্তিগত পছন্দ, আগ্রহ, নির্ভরযোগ্যতা, আচরণ, অভ্যাস, অবস্থান বা গতিবিধি সম্পর্কিত দিকগুলি মূল্যায়ন, বিশ্লেষণ বা পূর্বাভাস প্রদান করে;

Sec 9(1): a child's data may be processed only with a parent or guardian's consent.
Sec 9(2): the child's rights and interests must be protected

THE SAME PATTERN, AGAIN AND AGAIN

Phone numbers leak through shops, apps and registration processes across South Asia


The Daily Star

বাংলা

Update

Pathao says cyber attack exposed users' names, emails and phone numbers

Published : 8 October 2026 • Updated : 9 October 2026, 14:20

SHARE

• TECHNOLOGY

Hackers claim ~19M records; the company confirms names, emails and phone numbers were accessed


The Daily Star

বাংলা

No bar for now on two telcos' value-added services

Published : 22 December 2020 • Updated : 23 December 2020, 02:23

SHARE

• BUSINESS

Regulators have scrutinised silent value-added billing.


Privacy, Security, and Surveillance in the Global South: A Study of Biometric Mobile SIM Registration in Bangladesh

Syed Ishtiaque Ahmed[1], Md. Romael Hoque[2], Shion Guha[3], Md. Rashidujjaman Rifat[4], Nicola Dell[5]

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TIME

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WORLD +

Men in India Are Buying Lists of Women's Phone Numbers in Order to Sexually Harass Them

ADD TIME ON GOOGLE

Customers wait to recharge their mobile phones at a mobile-phone store in Mumbai on Aug. 12, 2014 *Dhiraj Singh—Bloomberg/Getty Images*

by Feliz Solomon

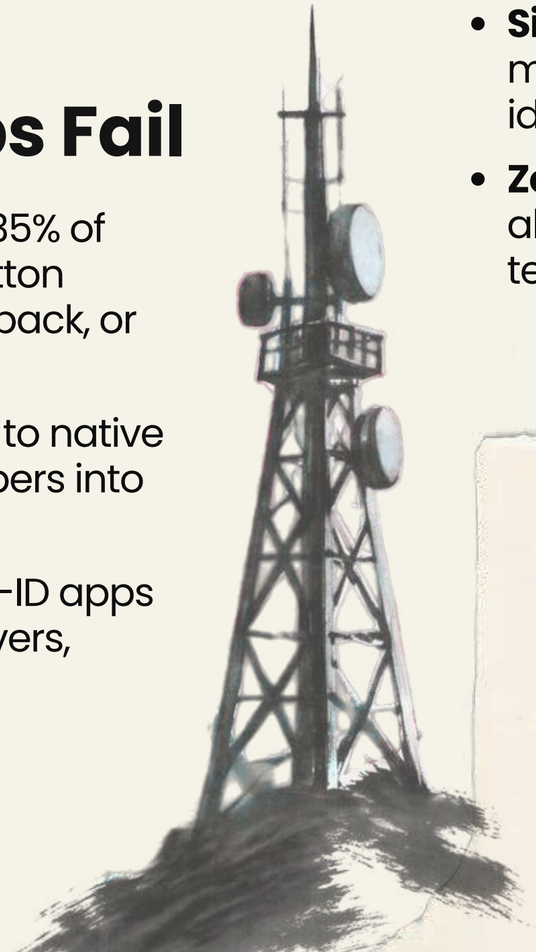
The same harassment pattern in India

WHY AN APP CAN'T DO THIS, AND THE NETWORK CAN

THE APPLICATION LAYER LIMIT

Why Smartphone Apps Fail

- **Blind to 20M+ Feature Phones:** About 35% of NTL's subscribers (20M+) use basic button handsets with zero app store, internet pack, or OS permissions.
- **Cellular Dialing Leak:** Apps hand calls to native phone dialers, exposing genuine numbers into riders' personal call logs permanently.
- **Data Leaves the Country:** Some caller-ID apps upload contact books to overseas servers, outside NTL's control.



THE NETWORK LAYER UTILITY

Why Telecom Core Dominates

- **Reaches Every Handset:** Operates natively across USSD (*1218#), SIM toolkit, and voice switches with zero data pack required.
- **Signaling-Level Routing:** Core switches bridge and mask calls before handsets ring, keeping genuine SIM identities completely hidden.
- **Zero Contact Harvesting:** No contact book scraping; all verification stays in Bangladesh on sovereign telecom signaling infrastructure.

NATIVE CARRIER ENFORCEMENT

Calls are bridged at the IMS signaling layer, before audio reaches any device

CONSUMER & RETAIL PRIVACY PRODUCTS

(PROTECTING EVERYDAY CITIZENS ON BUTTON PHONES AND SMARTPHONES)



YOUR NUMBER STAYS HIDDEN

Core IMS proxy generates disposable 24-hour virtual numbers for e-commerce deliveries.

TWO WAYS TO ACTIVATE

Consumer USSD (*1218*3#)

Any customer can dial *1218*3# to get a temporary burner number valid for one day.

B2B webhook partnership

- **Parcel Label Masking:** E-commerce checkouts auto-print temporary numbers directly onto boxes.
- **Normal Rider Dialing:** Couriers call normally from standard phone dialers; no app required.

CORE NETWORK BRIDGING

Network-Level Routing: Runs inside NTL core switches, without depending on third-party apps

⬇

Instant Memory Lookup: Translates the proxy to your real SIM in milliseconds upon dialing.

⬇

Two-Way Number Masking: Neither side's call log shows the other's real number.

⬇

Cross-Carrier Support: Works for calls from any operator

EXPIRY & LEGAL ESCROW

- **24-Hour Auto-Burn:** Proxy link dissolves permanently once delivered or after 24 hours.
- **Efficient Number Reuse:** Numbers return to the shared pool for continuous reuse.
- **Warrant Disclosure:** The link between the temporary and real number is revealed only on a High Court warrant, with two NTL officers approving
- **Anti-Crime Guard:** Defuses street harassment without creating an untraceable criminal haven.

TOP UP WITHOUT WRITING IN PAPER LEDGERS

A one-time code at the counter: the shopkeeper never sees your number.

SOLUTION 1 · LOCAL COUNTER RECHARGE

DYNAMIC COUNTER TOKEN (*1218*2#)

- **Zero Spoken Numbers:** Customer dials *1218*2#; retailer enters only the 6-digit token into his terminal.
- **Shorter to Type:** Six digits are quicker than eleven, so counters move faster.
- **Fewer Misdial Disputes:** A short token reduces mistyped numbers and the reversals that follow

SOLUTION 2 · REMOTE FAMILY RECHARGE

POSSIBLE EXTENSION

- **Permanent Family Alias:** Register an 8-digit alias PIN or QR code for dependent family members.
- **Remote Village Privacy:** Relatives in Dhaka can top up family lines in rural villages without exposing real phone numbers
- **Protects Vulnerable Users:** Shields rural women, elderly parents, and young students from opportunistic phone number harvesting.

CONSUMER PRIVACY & IDENTITY | RECYCLED NUMBER RISKS

A RECYCLED NUMBER CAN HAND A STRANGER HIS BANK OTPS

STAGE 1

Dormancy Period

Inactive SIM reaches BTRC's mandatory 12-month regulatory cutoff for cancellation and recycling.

Status: BTRC Regulatory Cutoff

STAGE 2

Quarantine Buffer

The number is held out of circulation (quarantine) while banks are notified

Status: Routing Isolation

STAGE 3

CAMARA B2B Push

Automated webhook notifies commercial partner banks and bKash to unlink dormant profiles immediately.

Status: Bank Webhook Delink

STAGE 4

Clean Reissue

New subscriber receives a pristine number completely unburdened by legacy financial alerts or debts.

Status: Pristine Line Activated

Automated GSMA CAMARA webhooks notify partner banks to delink stale accounts before physical SIM reissue.

CUSTOMER BENEFIT

Tanvir's bank accounts, MFS wallets, and private messages are protected from being hijacked by the next buyer of his lapsed number. (Severing the Ghost Identity)

THE GHOST IDENTITY

Under BTRC rules, dormant numbers are recycled after 12 months. When reissued, the new buyer inherits unlinked bKash, bank, and social media OTPs of the previous owner.

NTL ENTERPRISE VALUE

Protects 1.2M elite post-paid subscribers (+BDT 800/mo ARPU) while monetizing verified B2B delink webhooks (upside, not in our numbers) with partner banks. (Churn & Monetization)

ONE DIAL TAKES A CHILD'S LINE OUT OF PROFILING

Default zero-profiling and guardian line shield operationalizing PDPA Section 9 & Definition 2(13).



STATUTORY SECTION 9 COMPLIANCE

PDPA Sec 9 requires guardian consent for a child's data; the case brief bans profiling and tracking of minor

PARENTAL EMPOWERMENT VIA USSD

Parental USSD Dial (*1218*4#)

- **Shield Activation:** Dialing *1218 *4# immediately shields shared family handsets across on any of NTL's 58.6M lines with zero smartphone app required.
- **Network-Layer Enforcement:** Core switches drop third-party ad-tracking beacons and block unauthorized third-party charges

ZERO-PROFILING STATUTORY COMPLIANCE

- **No Behavioral Guesswork:** Commercial AI models never ingest line telemetry without affirmative guardian consent, avoiding unlawful child surveillance.
- **Compliance Support:** Helps NTL meet PDPA Section 9 and Definition 2(13) and reduces exposure to penalties under the Act.

B2B FINTECH & LEGAL INFRASTRUCTURE

(ENTERPRISE MONETIZATION, BANKING TRUST, AND REGULATORY HARMONY)



PDPA SAYS ERASE.

BTRC SAYS KEEP.

How NTL resolves the legal conflict between privacy rights and telecom record retention

COMMERCIAL ERASURE LAYER

PRIVACY LAW · PDPA 2026 (SECTION 13(2))

- **One-Dial Deletion (*1218*6#):** Customers dial *1218*6# on any phone to demand complete deletion of their marketing data.
- **24-Hour Full Purge:** Advertising profiles, app activity logs, and customer service tags are wiped within 24 hours.
- **Backup Key Destruction:** Digital encryption keys are permanently destroyed, making old server backups completely unreadable.
- **Locked Down:** Write-once storage, no link to commercial systems, and every unlock is logged.

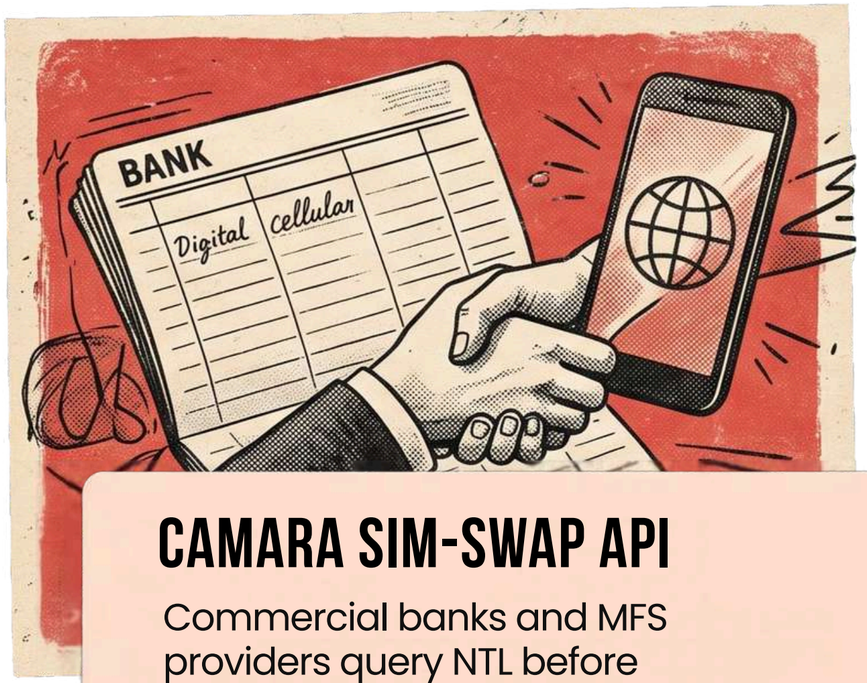
STATUTORY RETENTION VAULT

TELECOM ACT 2001 (SECTION 97)

- **24-Month Legal Hold:** Phone networks are legally required to keep basic call timestamps and tower records for 24 months.
- **Physical Clean Wall:** These records are locked in an isolated vault physically cut off from all marketing databases.
- **High Court Warrant Only:** Records open only on a High Court warrant, and only when two NTL officers (the Chief Legal Officer and the DPO) both approve.
- **Zero Rogue Access**: Neither company employees nor external hackers can view, alter, or leak preserved call logs.

BANKS ASK ONE QUESTION AND GET ONE WORD

Monetizing SIM-swap verification for commercial banks



CAMARA SIM-SWAP API

Commercial banks and MFS providers query NTL before processing high-value financial transfers: “Was this SIM swapped recently?”

NTL returns a boolean True/False.

EXECUTION PROTOCOL

- **Stops Account Takeover Fraud:** Banks can detect if a criminal hijacked a customer's SIM before approving cash transfers.
- **Only a Yes/No Leaves NTL:** No names, balances or call logs.
- **Invisible to legitimate customers:** extra checks apply only when a swap is detected

Commercial Bank	← QUERY →	NTL Core
"Was this SIM swapped in the last 48 hours?"		
Commercial Bank	← ANSWER ←	"FALSE"

Proven Precedents: Dialog Axiata Sri Lanka CAMARA Deployment; Telefónica Brazil Open Gateway.

SOVEREIGN AI & CYBER DEFENSE

(LOCAL LLM SCRUBBING, ANTI-DEEPFAKE, AND TRANSPARENT EXPLAINABILITY)



THE AI NEVER SEES WHO YOU ARE



In-memory data scrubbing protect customer identities across all AI tools

BEFORE THE AI SEES THE QUERY

LOCAL PRIVACY SHIELD

- **Local PII Stripping:** Customer IDs and phone numbers are removed locally inside Bangladesh.
- **Prompt Injection Defense:** Malicious tricks and manipulated prompts are filtered before reaching the AI.
- **Temporary Placeholders Only**: Identity tags vanish from memory as soon as the query is answered.

Request → **Scrub** → **AI**

WHEN THE AI RESPONDS

ZERO DATA EXPORT · SECTION 29

- **Data Stays Home:** Only scrubbed text can leave. Identifiers and restricted data stay in Bangladesh.
- **Never Trains Public AI:** Customer details are never saved or absorbed into public AI models.
- **Clear Bangla Receipts:** Automated decisions are sent back to customers as simple, transparent SMS cards.

STOPPING AI ATTACKERS, DEEPFAKES & MODEL INVERSION

Three practical defenses against AI-assisted fraud and data extraction

DEEPFAKE & BIOMETRIC DEFENSE

- **Active 3D Liveness:** Liveness check at e-KYC to resist replayed videos and photos.
- **Zero Biometric Retention:** Facial geometry is checked only in volatile memory; no photos are saved to disk.

MODEL INVERSION DEFENSE

- **Privacy-Protected Training:** Where a model is fine-tuned, calibrated noise is added in training so single records can't be recovered.
- **Leakage Testing:** We test models for memorised records before release

DATA EXFILTRATION DEFENSE

- **Traffic Anomaly Monitoring:** Continuous surveillance flags abnormal, high-frequency database queries across all APIs.
- **Automatic Session Cut-Off:** Compromised developer tokens and rogue requests are severed automatically without human delay.

TESTED BEFORE DEPLOYMENT.

No model goes live untested, and every automated decision comes with a reason in plain Bangla

AUTOMATED RISK GATE

PRE-LAUNCH SAFETY GATE

- **Verified Training Consent:** Models are blocked from production unless training data proves lawful consent (PDPA Sec 16).
- **No Unapproved Models:** Unapproved algorithms across churn, fraud, or marketing cannot touch live subscriber data.

1. Model Submission

⬇

2. Automated Privacy & Bias Scan

⬇

3. Pass or Deployment Frozen

EXPLAINED IN PLAIN BANGLA.

CUSTOMER EXPLANATIONS

STATUTORY TRANSPARENCY (PDPA SEC 15)

- **Plain-Language Reasons:** Customers receive plain-language Bangla reasons whenever AI adjusts credit limits or services.
- **One-Dial Human Appeal:** Subscribers can contest any automated decision and request human review via *1218#.

SMS:

আপনার ক্রেডিট লিমিট পুনর্মূল্যায়ন করা হয়েছে।
কারণ: গত ৩ মাস নিয়মিত বিল পরিশোধ।
বিস্তারিত জানতে ডায়াল করুন *1218#

Answers management Questions 6 (explainability) and 8 (accountability)

ENTERPRISE GOVERNANCE & STRATEGIC TRUST

(INTERNAL DEFENSES, STATUTORY TRUTH TABLE, EXECUTION, AND BOARD COVENANTS)



PROTECTING EMPLOYEE DATA. BLOCKING SHADOW AI.

Protecting staff records, stopping data leaks to public AI tools, and training people for their roles.

PILLAR 1 · HR ISOLATION

HR DATA FIREWALLS & ISOLATION

- **Cryptographic Separation**: Salaries, medical claims, and appraisal records are strictly walled off from commercial databases.
- **Zero Marketing Co-Mingling:** Prevents staff identity files from ever entering customer targeting or analytics pipelines.

Supports PDPA Section 16 (Duty of Confidentiality)

PILLAR 2 · SHADOW AI SHIELD

SANITIZED ENTERPRISE AI PORTAL

- **Private AI Workspace:** Provides employees an approved, access-controlled AI workspace on NTL's private cloud for everyday corporate productivity.
- **Zero Public Leakage:** Gives staff an approved tool, and the egress filter blocks unapproved AI sites.

PILLAR 3 · CONTINUOUS DRILLS

INSIDER DEFENSE & AWARENESS DRILLS

- **Simulated Bribe & Phishing Tests:** Monthly unannounced social engineering drills train call-center agents against external bribes.
- **Executive Compensation Link:** Proposed: departmental privacy scores feed into management performance reviews

EVERY PDPA DUTY HAS AN OWNER AND A PROOF

Section references per PDPA 2026 (Act No. 63 of 2026). Each duty has a control and a proof

STATUTORY DUTY	PDPA SECTION	NTL PRIVACY CONTROL	VERIFIABLE AUDIT PROOF
Child Protection	**Sec 9 & Def. 2(13)**	Guardian Child Shield via *1218#	Parental Consent Record & Zero-Ad Logs
Right to Erasure	**Sec 13(2)**	Commercial Profile 24-Hour Deletion	Automated Deletion Certificate
Data Confidentiality	**Sec 16**	Local AI Identity Scrubbing	De-Identified Query Log (no raw data kept)
Chief Data Officer	**Sec 23**	Independent Board-Level CDO	Board Governance Charter & Release Veto (NTL policy)
Lawful Interception	**Sec 24**	Isolated Jashore Compliance Vault	Dual-Key Judicial Warrant Log
Cross-Border Transfer	**Sec 29**	National Data Residency Lock	Outbound Network Boundary Drop Logs

FOUR PHASED MILESTONES

24 months in four phases. Phase 2 money is released only if the pilot succeeds.

PILOT

CORE PRIVACY & KIOSK PILOT

- **Sovereign Gateway:** Deploys on-premise AI data-scrubbing nodes.
- **25k Kiosk Pilot:** Tests Blind Flexiload across selected districts.
- **Airtime Vault:** Activates PIN-protected balance shield via USSD.

MASS

NATIONWIDE RETAIL & VOICE

- **250k Safe Shops:** Removes customer numbers from shop ledgers.
- **Dialect Voice IVR:** Spoken consent for keypad and rural users.
- **Child Shield:** One-dial parental protection active nationwide.

B2B

FINTECH APIS & COMMERCE

- **Bank SIM-Swap APIs**: Commercial launch with top banks and MFS.
- **Delivery Proxy:** Anonymous number routing for parcel couriers.

MATURITY

AUDIT PROOF & STEADY STATE

- **1-Click Audit Pack:** Generates instant compliance proof for regulators.
- **Self-Funding OpEx:** Savings and partner revenue cover running costs (modeled).
- **Zero Downtime:** Rolled out in stages, without interrupting service

THE "NEVER LIST": FIVE BINDING PROMISES

Non-negotiable board covenants backing customer privacy with legal accountability

NEVER PERMIT SILENT CHARGES

Airtime is locked; third-party billing requires a PIN.

NEVER PROFILE CHILDREN

Minor lines are completely excluded from AI tracking.

NEVER EXPORT DATA OVERSEAS:

Citizen records remain strictly on sovereign soil (Sec 29).

NEVER SELL LOCATION DATA

Tower coordinates are never shared with commercial brokers.

ENACTED IN NTL BOARD CHARTER

NEVER PROFIT FROM BREACHES:

Guaranteed free SIM replacement and automatic breach refunds.